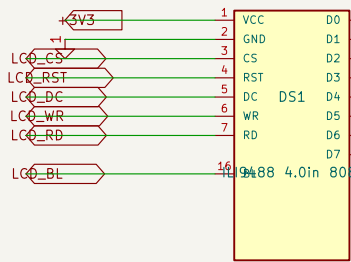


DISPLAY – ILI9488 4.0in 320x480

8–bit 8080 parallel interface (mandatory for SNES emulation speed)

Control signals:

GPI012 / FPC9
GPI013 / FPC15
GPI014 / FPC10
GPI046 / FPC11
+3V3 (FPC12, RO)
+3V3 (FPC33, BL)

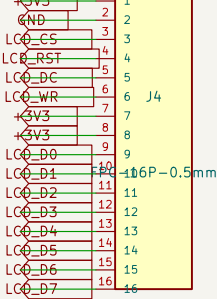


8–bit data bus:

GPI04 (FPC17)
GPI05 (FPC18)
GPI06 (FPC19)
GPI07 (FPC20)
GPI08 (FPC21)
GPI09 (FPC22)
GPI010 (FPC23)
GPI011 (FPC24)

FPC RIBBON CONNECTOR (J4)

(40P on PCB, 16 active signals shown)



ILI9488 Panel FPC–40P Pin Mapping (source: components.md):

Panel 6,7=VDDI/VDDA(+3V3) 9=CS 10=DC 11=WR 12=RD 15=RESET

Panel 17–24=DB0–DB7 33=LED–A(BL) 34–36=LED–K(GND) 38,39=IM0,IM1(+3V3) 40=IM2(GND)

Interface mode: IM2=0 IM1=1 IM0=1 = 8080 8–bit parallel

- GPIO4–11 form contiguous 8–bit bus for efficient register–level DMA
- WR strobes data on rising edge, RD tied HIGH (+3V3, write–only mode)

□ FPC PIN REVERSAL on PCB: panel pin N ↔ J4 pad (41–N).

e.g. panel pin 17 (DB0) contacts J4 pad 24. See datasheet_specs.py for the connector–side mapping.

Generated by scripts/generate_schematics
ESP32 Emu Turbo – Handheld Retro Console

Sheet: /
File: 03–display.kicad_sch

Title: Display – ILI9488 4.0in 8080 Parallel

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